



Application Note

Ensemble E4, E6, E8 Differences From E1, E3, E5, E7

Version 1.0

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Overview

The Ensemble E4, E6, and E8 devices have some functional differences compared to the E1, E3, E5, and E7 devices. This document outlines the items that a designer should be aware of when doing a design for the E4, E6, or E8 devices and which ones are taken care of when using the Alif-provided Device Family Pack (DFP) drivers.

Functional Differences

- SRAM1 is now at 0x0240_0000 instead of 0x0800_0000
 - For existing projects, the linker and device tree will need modifications
- SRAM6-9 are removed
 - RAM available at locations starting with 0x6 is not available
 - Instead, use the additional 1.5MB in SRAM1
- HCLK and PCLK are now derived from SYSPLL, not ACLK, so HCLK and PCLK will not slow when the ACLK divider is increased.
 - When configuring HCLK and PCLK frequencies, use new calculations for the divider as provided in the HWRM.
- There are several new CGU clock gates in the CLK_EN register. Most default to ON, a few like the CLK_76M8 going to I2S and PDM blocks default to off.
 - If not using the latest SDK, please check HWRM for clock enable requirements
- OSPI IP has been updated to 2.01, and has some configuration changes:
 - SSIC_XIP_EXT_SSEL_EN 0->1, meaning the XIP_SER register no longer exists, and there is no manual control of slave select lines. Access to XIP_SER will fault. Slave selection is now performed on the basis of bit 28 of the address, so 0xA/C0000000 selects slave 0, and 0xB/D0000000 selects slave 1.
 - When working with multiple slave devices connected to the same OSPI instance, use a designated address window to select between them
 - XIP_WRITE_CTRL now has a new bit XIPWR_DFS_HC, and hard-coded DFS support now works for writes as well as reads. This defaults to on.
 - When working with external RAM, use these new registers to improve compatibility
 - Unaligned, narrow, and sparse writes to DDR memory now work correctly, due to upgrades of the SSI, and the insertion of an XHB-500 bridge, as long as XIP_WRITE_CTRL.XIPWR_DFS_HC and XIP_CTRL.DFS_HC are both set.
 - Use of external RAM as general-purpose memory and shared memory between multiple bus masters is now supported
- The function indexes of pin P5_6 have shifted down by 1.
 - When configuring pin mux for pin P5_6 use the updated definition for the new device

Security/firewalling notes

- Extra memory aliases exist, such as OSPI0 at 0x2000_0000. These could defeat the pre-existing firewall configuration. All of them can be disabled - the intent is that this should be possible via SERAM device config JSON settings.
 - This excludes M55.
 - When working with external memory over OSPI, ensure your firewall configuration covers the new address in the 0x2000_0000 range.

Changes Covered by Alif drivers

- I2C has a configuration option IC_EMPTYFIFO_HOLD_MASTER_EN, which means STOP and RESTART indications must be written into the transmit FIFO.
 - Additional I2C capabilities - user can decide whether a STOP or RESTART condition is issued as the transaction goes on
- OSPI clock gating that was in the MRAM controller register spacing is now in CFGMST0 at 0x4902_F03C. Again, not in the Bolt HWRM. The gating is not enabled by default, different from Bolt.
 - When not using Alif SDK (CMSIS or Zephyr) use the new clock enable register for OSPI at 0x4902_F03C

Other

- Performance will be noticeably different, mainly from things being faster. No significant performance losses are expected. Differences will be:
 - Faster MRAM
 - Full duplex SRAM access (combined read+write to 1 SRAM could not exceed 3.2GB/s in Ensemble E1, E3, E5 and E7 - it can approach 6.4GB/s in Ensemble E4, E6, and E8).
 - Faster M55->M55 TCM access
 - Possibly slightly higher SRAM latency due to 200MHz clock
 - Higher SPI latency due to clock domain crossing
 - Higher power when running faster
- Access to addresses that belong to clock-gated areas will result in a bus fault, whereas on earlier devices, you just read zeroes.
- The registers for the following are at different addresses. Please refer to the HWRM
 - ADC
 - DAC
 - CMP
 - LPCMP
 - BOD / BOR

Document History

Version	Change Log
1.0	initial release to the public should be version 1.0